Scattered vs. Interpolated Lagrangian Data: Rise Velocity and Circularity Accuracy

Short Summary

Supplementary material

1 Objective

We assess whether Lagrangian particle tracking (LPT) data should be used as-is (scattered) or first re-interpolated onto the Eulerian grid when computing two bubble diagnostics: *rise velocity* and *circularity*. Additionally, we evaluate whether interpolation schemes are the dominant source of errors. The test case is a 2D axisymmetric rising bubble (Re = 100, Bo = 100, $D = 0.63\,\mathrm{m}$, 450 snapshots). For the Lagrangian tracking, two different numerical interpolation schemes were tested. Scheme 1 is linear for both velocity field and gas volume fraction while scheme 2 is cubic method for both fields.

2 Definitions

Rise velocity is computed two ways:

- *Centroid-based*: finite-difference the α_g-weighted centroid position. Uses only α_g.
- *Volume-averaged*: α_g-weighted mean of V_z. Uses both α_g and V_z.

Circularity $C = 4\pi A/P^2$ ($C = 1$ for a circle):

- *Grid-based* (Eulerian & re-interpolated data): iso-contour at $\alpha_\mathrm{g} = 0.5$, half-domain mirrored for axisymmetry.
- *Scattered*: threshold particles at $\alpha_\mathrm{g} \geq 0.5$, mirror across the axis.

Data representations:

- *Approach A* — use scattered particle data directly.
- *Approach B* — re-interpolate scattered data onto the Eulerian grid (natural-neighbour interpolation), then apply the same grid-based formulas as the Eulerian reference.

3 Key Findings

(1) **Rise velocity is poorly approximated from scattered data and improves dramatically after re-interpolation**. Both the centroid-based and volume-averaged definitions yield comparable accuracy; the difference between the two methods is small relative to the overall improvement gained by re-interpolation. (see figures 1 and 2)

(2) **Circularity is the diagnostic most affected by the scattered representation.** Non-uniform particle clustering distorts the boundary extraction, producing large deviations from the Eulerian reference—especially during the initial transient. Re-interpolation onto the Eulerian grid eliminates this bias almost entirely by restoring the spatial regularity needed for iso-contour extraction.

(3) **High order interpolation schemes such as cubic slightly improve the Re-interpolated results while they still remain inaccurate (see figures 3 and 2).**

Recommendation: Scattered representation of the Lagrangian data is found to be the major source of error while interpolation scheme comes second as the main cause of inaccuracies in rise

characteristics. Thus, the discussion recommends re-interpolating scattered Lagrangian data onto a fixed grid before computing integral diagnostics, particularly for shape-based metrics such as circularity. Meanwhile, high order interpolation scheme improves the accuracies.

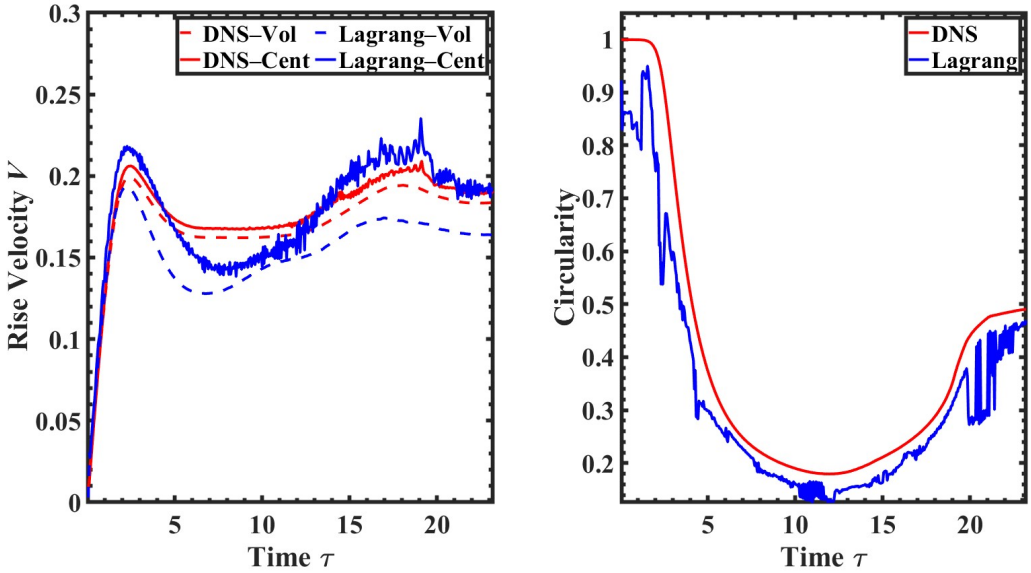

Figure 1: Scattered Lagrangian data (Approach A, scheme 1). Left: rise velocity (solid = centroid, dashed = vol-avg; red = Eulerian, blue = Lagrangian). Right: circularity.

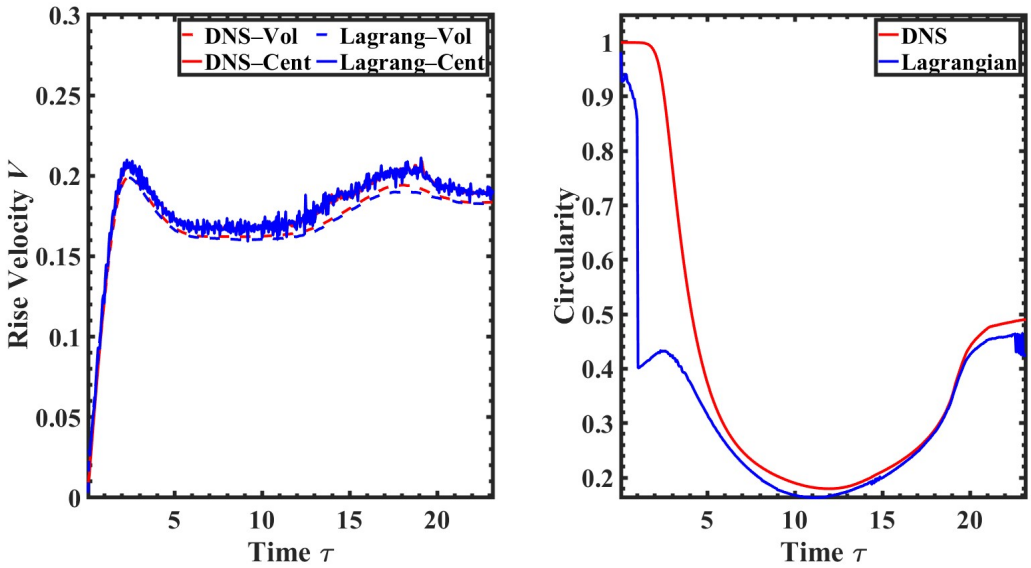

Figure 2: Re-interpolated Lagrangian data (Approach B, scheme 1). Left: rise velocity. Right: circularity. Both Lagrangian curves (blue) now closely track the Eulerian reference (red).

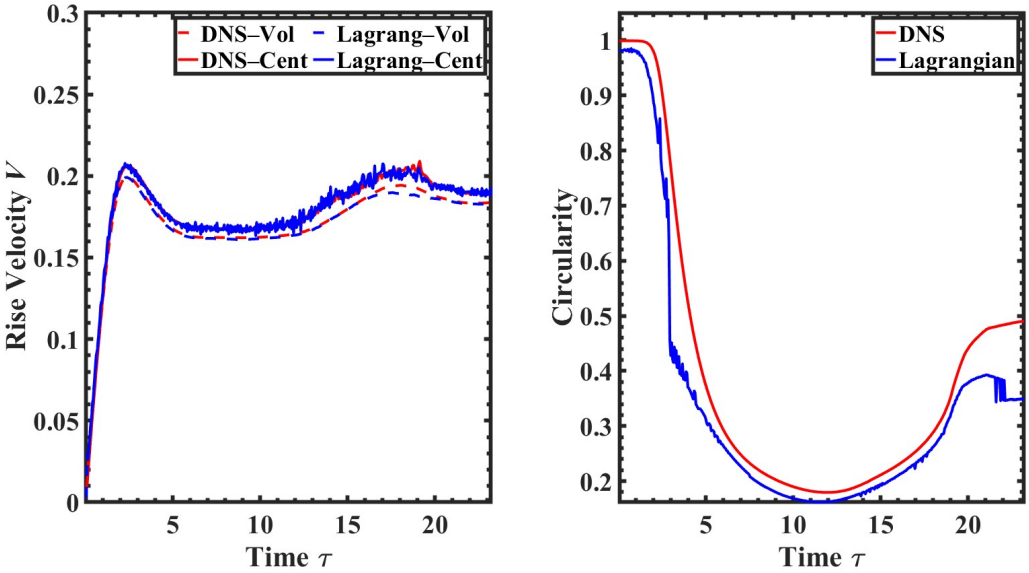

Figure 3: Re-interpolated Lagrangian data (Approach B, scheme 2). Left: rise velocity. Right: circularity. Both Lagrangian curves (blue) now closely track the Eulerian reference (red).